essentially what happens inside a television—they found something remarkable. A pattern very much akin to that of Figure 4.8 emerged. Their experiment therefore showed that electrons exhibit interference phenomena, the telltale sign of waves. At dark spots on the phosphorescent screen, electrons were somehow “canceling each other out” just like the overlapping peak and trough of water waves. Even if the beam of fired electrons was “thinned” so that, for instance, only one electron was emitted every ten seconds, the individual electrons still built up the bright and dark bands—one spot at a time. Somehow, as with photons, individual electrons “interfere” with themselves in the sense that individual electrons, over time, reconstruct the interference pattern associated with waves. We are inescapably forced to conclude that each electron embodies a wave-like character in conjunction with its more familiar depiction as a particle.

Although we have described this in the case of electrons, similar experiments lead to the conclusion that all matter has a wave-like character. But how does this jibe with our real-world experience of matter as being solid and sturdy, and in no way wave-like? Well, de Broglie set down a formula for the wavelength of matter waves, and it shows that the wavelength is proportional to Planck’s constant ħ. (More precisely, the wavelength is given by ħ divided by the material body’s momentum.) Since ħ is so small, the resulting wavelengths are similarly minuscule compared with everyday scales. This is why the wave-like character of matter becomes directly apparent only upon careful microscopic investigation. just as the large value of c, the speed of light, obscures much of the true nature of space and time, the smallness of ħ obscures the wave-like aspects of matter in the day-to-day world.

Waves of What?

The interference phenomenon found by Davisson and Germer made the wave-like nature of electrons tangibly evident. But waves of what? One early suggestion made by Austrian physicist Erwin Schrödinger was that the waves were “smeared-out” electrons. This captured some of the “feeling” of an electron wave, but it was too rough. When you smear something out, part of it is here and part of it is there. However, one never encounters half of an electron or a third of an electron or any other fraction, for that matter. This makes it hard